emerged from the quantum foam and froth of the big bang singularity. We may learn for sure the meaning or the meaninglessness of the oft-asked question, “What preceded the big bang?” We may learn for sure whether quantum foam produces multiple universes with ease, and the full details of how spacetime gets destroyed in the singularity at the core of a black hole or in the big crunch, and how and whether and where spacetime gets re-created again. And we may learn whether the laws of quantum gravity permit or forbid time machines: Must time machines always self-destruct at the moment they are activated?

- The laws of quantum gravity are not the final set of physical laws along the route that has led from Newton to special relativity, to general relativity and quantum theory, and then to quantum gravity. The quantum gravity laws will still have to be married to (unified with) the laws that govern nature’s other fundamental forces: the electromagnetic force, the weak force, and the strong force. We will probably learn the details of that unification in the coming century—and again, most likely sooner rather than later; and that unification may radically alter our view of the Universe. And what then? No human today can foresee beyond that point, I believe—and yet, that point may well come in my own lifetime, and in yours.

In Closing, November 1993

Albert Einstein spent most of his last twenty-five years in a fruitless quest to unify his general relativistic laws of physics with Maxwell’s laws of electromagnetism; he did not know that the most important unification is with quantum mechanics. He died in Princeton, New Jersey, in 1955 at the age of seventy-six.

Subrahmanyan Chandrasekhar, now eighty-three years old, continues to plumb the secrets of Einstein’s field equation, often in collaboration with much younger colleagues. In recent years he has taught us much about pulsations of stars and collisions of gravitational waves.

Fritz Zwicky became less a theorist and more an observational astronomer as he aged; and he continued to generate controversial, prescient ideas, though not on the topics of this book. He retired from his Caltech professorship in 1968 and moved to Switzerland, where he spent his final